

DD_ASR-02 - Information Request - Developer Implementation Guide

Version: 1.0 Status: Draft for review Bundle: 10_other Audience: mid-level backend developers, ticketing developers, QA
Parent DD: DD_ASR-02 - Information Request

1. How To Read This Guide

Information request is a TCK ticket flow. It is not a separate ASR case engine.

The flow creates a TCK ticket, resolves guidance from ASR config, optionally performs research, then stores a customer-visible answer and resolves the ticket.

2. Step 1 - Seed TCK Information Categories

Seed TCK categories.

ASR answer templates are canned agent-response content. They are not NOT-01 notification templates. If an agent chooses to send the answer through SMS, email, WhatsApp, push, or portal notification, the delivery call must go through NOT-01; NOT-01 owns template selection, channel, language, fallback, and retry policy.

```
insert into ticket_category_config (  
    category_id,  
    operator_code,  
    type_code,  
    category_code,  
    display_name,  
    default_priority,  
    default_queue_code,  
    default_sla_policy,  
    wo_allowed,  
    default_wo_kind,  
    status  
) values  
(  
    'tcat-package-info', 'WIK', 'PACKAGE_OR_CATALOG_QUERY', 'PACKAGE_EXPLANATION', 'Package explanation', 'P4', 'CARE_L1',  
    'tcat-coverage-info', 'WIK', 'PACKAGE_OR_CATALOG_QUERY', 'COVERAGE_INFORMATION', 'Coverage information', 'P4', 'CARE_L1',  
    'tcat-invoice-info', 'WIK', 'BILLING_COMPLAINT', 'INVOICE_EXPLANATION', 'Invoice explanation', 'P3', 'BILLING_CARE', 'I',  
    'tcat-kyc-info', 'WIK', 'KYC_SUPPORT', 'KYC_STATUS_INFORMATION', 'KYC status information', 'P4', 'CARE_L1', 'INFO_P4_48'
```

These are TCK categories. ASR-02 only adds guidance and answer rules.

3. Step 2 - Seed ASR Info Category Config

```
insert into asr_info_category_config (  
    config_id,  
    operator_code,  
    category_code,  
    default_type_code,  
    answer_source_module,  
    research_required,  
    default_queue_code,  
    selfcare_allowed,  
    answer_template_code,  
    conversion_policy_json,  
    status  
) values  
(  
    'airc-package-info', 'WIK', 'PACKAGE_EXPLANATION', 'PACKAGE_OR_CATALOG_QUERY', 'SIP-01', false, 'CARE_L1', true, 'PACKA',  
    'airc-coverage-info', 'WIK', 'COVERAGE_INFORMATION', 'PACKAGE_OR_CATALOG_QUERY', 'RLM-CFG-01', false, 'CARE_L1', true,  
    'airc-invoice-info', 'WIK', 'INVOICE_EXPLANATION', 'BILLING_COMPLAINT', 'BIL-02-READ-01', true, 'BILLING_CARE', true, '  
    'airc-kyc-info', 'WIK', 'KYC_STATUS_INFORMATION', 'KYC_SUPPORT', 'ILM-CFG-01', false, 'CARE_L1', true, 'KYC_STATUS_BASI'
```

4. Step 3 - Seed Answer Templates

```
insert into asr_info_answer_template (  
    template_id,  
    operator_code,
```

```

    template_code,
    language_code,
    subject_template,
    body_template,
    required_variables_json,
    status
) values (
    'ait-package-basic-en',
    'WIK',
    'PACKAGE_EXPLANATION_BASIC',
    'en',
    'Your current package information',
    'Your current package is {{packageName}}. The current monthly recurring charge is {{monthlyPrice}} {{currency}} before a
    ["packageName","monthlyPrice","currency"]',
    'ACTIVE'
);

```

Template rendering must fail safely if required variables are missing. Do not send an answer with unresolved placeholders.

5. Step 4 - Customer Creates Information Request

Self-care or Backoffice calls TCK:

```

POST /api/tickets
Idempotency-Key: selfcare-info-20260603-001
Content-Type: application/json

{
  "operatorCode": "WIK",
  "typeCode": "PACKAGE_OR_CATALOG_QUERY",
  "categoryCode": "PACKAGE_EXPLANATION",
  "sourceChannel": "SELF_CARE",
  "customerId": "CUS-100045",
  "accountId": "ACC-900045",
  "subscriptionId": "SUB-700045",
  "subject": "Package information request",
  "description": "Customer wants to know current package and price.",
  "customerVisible": true
}

```

Before allowing self-care create, validate:

```

select selfcare_allowed
from asr_info_category_config
where operator_code = 'WIK'
and category_code = 'PACKAGE_EXPLANATION'
and status = 'ACTIVE';

```

If selfcare_allowed = false, return:

```

403 ASR_INFO_SELF_Care_CATEGORY_NOT_ALLOWED

```

6. Step 5 - Load Guidance

Backoffice opens the ticket:

```

GET /api/asr/information-requests/tck-20260603-000020/guidance

```

Response:

```

{
  "ticketId": "tck-20260603-000020",
  "categoryCode": "PACKAGE_EXPLANATION",
  "answerSourceModule": "SIP-01",
  "researchRequired": false,
  "answerTemplateCode": "PACKAGE_EXPLANATION_BASIC",
  "allowedActions": [
    "ANSWER_AND_RESOLVE",
    "CREATE_RESEARCH_TASK",
    "CONVERT_TO_COMPLAINT"
  ]
}

```

Frontend uses this to show the right actions. Do not hardcode topic behavior in the UI.

7. Step 6 - Fetch Source Data

For package explanation, the service gathers:

1. Subscription from SUB-LM.
2. Package version from SIP-01.
3. Service summary from PLM-CFG-01 if needed.

Example internal source refs:

```
[
  {"module": "SUB-LM", "ref": "SUB-700045"},
  {"module": "SIP-01", "ref": "pkg-fiber-100m"}
]
```

If source API is unavailable, create a research task or move ticket to WAITING_INTERNAL.

8. Step 7 - Create Research Task When Needed

Invoice explanation may need billing-care review.

```
POST /api/asr/information-requests/tck-20260603-000021/research-tasks
Content-Type: application/json
```

```
{
  "researchType": "TEAM_REVIEW",
  "sourceModule": "BIL-02-READ-01",
  "assignedQueueCode": "BILLING_CARE",
  "requestPayload": {
    "invoiceId": "INV-2026-06-000045",
    "question": "Explain prorated line item."
  }
}
```

Insert:

```
insert into asr_info_research_task (
  research_task_id,
  ticket_id,
  operator_code,
  research_type,
  source_module,
  status,
  request_payload_json,
  assigned_queue_code,
  created_at
) values (
  'airt-001',
  'tck-20260603-000021',
  'WIK',
  'TEAM_REVIEW',
  'BIL-02-READ-01',
  'OPEN',
  '{"invoiceId":"INV-2026-06-000045","question":"Explain prorated line item."}',
  'BILLING_CARE',
  now()
);
```

Then update TCK ticket:

```
status_code = WAITING_INTERNAL
assigned_queue_code = BILLING_CARE
```

9. Step 8 - Complete Research Task

Billing-care user completes the research:

```
POST /api/asr/information-requests/research-tasks/airt-001/complete
Content-Type: application/json
```

```
{
  "resultPayload": {
    "explanation": "The line is a prorated charge for 10 days after package upgrade.",
    "sourceRefs": [
      {"module": "BIL-02-READ-01", "ref": "INV-2026-06-000045"}
    ]
  }
}
```

```
    ]
  }
}
```

Update research task:

```
update asr_info_research_task
set status = 'COMPLETED',
    result_payload_json = '{"explanation":"The line is a prorated charge for 10 days after package upgrade."}',
    completed_at = now()
where research_task_id = 'airt-001';
```

Then return ticket to the answering queue:

```
status_code = ASSIGNED
assigned_queue_code = CARE_L1 or prior owner
```

10. Step 9 - Answer And Resolve

CSR posts final answer:

```
POST /api/asr/information-requests/tck-20260603-000020/answer-and-resolve
Content-Type: application/json
```

```
{
  "templateCode": "PACKAGE_EXPLANATION_BASIC",
  "languageCode": "en",
  "answerBody": "Your current package is Fiber 100M. The monthly recurring charge is 4,500 KES before any active discounts.",
  "sourceRefs": [
    {"module": "SIP-01", "ref": "pkg-fiber-100m"},
    {"module": "SUB-LM", "ref": "SUB-700045"}
  ],
  "resolutionCode": "ANSWER_PROVIDED"
}
```

Insert answer:

```
insert into asr_info_answer (
  answer_id,
  ticket_id,
  operator_code,
  template_code,
  language_code,
  answer_body,
  source_refs_json,
  answered_by_user_id,
  answered_at
) values (
  'aia-001',
  'tck-20260603-000020',
  'WIK',
  'PACKAGE_EXPLANATION_BASIC',
  'en',
  'Your current package is Fiber 100M. The monthly recurring charge is 4,500 KES before any active discounts.',
  '[{"module":"SIP-01","ref":"pkg-fiber-100m"}, {"module":"SUB-LM","ref":"SUB-700045"}]',
  'usr-care-001',
  now()
);
```

Then:

1. Add TCK customer-visible comment with answer body.
2. Resolve ticket through TCK status API.
3. Emit InformationRequestAnswered.
4. Project event to CUST-INT-01.

11. Step 10 - Convert To Another Flow

If a customer asks for invoice explanation and then says the charge is wrong, convert or link to complaint/adjustment path.

```
POST /api/asr/information-requests/tck-20260603-000021/convert
Content-Type: application/json
```

```
{
  "targetFlow": "ASR-03",
```

```

    "targetCategoryCode": "BILLING_DISPUTE",
    "reason": "Customer rejects invoice explanation and requests correction."
  }
}

```

Recommended implementation:

1. Create linked TCK ticket for ASR-03, or change current ticket category if policy allows.
2. Add `ticket_link` between old and new ticket.
3. Close original with `resolutionCode = CONVERTED`.
4. Emit `InformationRequestConverted`.

Never delete the original information-request history.

12. Suggested Worker Commands

These run inside TCK/ASR workers and can be consolidated.

```

asr-info:research-sla-scan
asr-info:template-validation-scan
asr-info:source-api-retry

```

Recommended schedule:

Command	Frequency
<code>asr-info:research-sla-scan</code>	Every 10 minutes.
<code>asr-info:template-validation-scan</code>	On template change and nightly.
<code>asr-info:source-api-retry</code>	Every 5 minutes for failed API research tasks.

13. Error Codes

Error code	Meaning
<code>ASR_INFO_CATEGORY_NOT_FOUND</code>	No active config for category.
<code>ASR_INFO_SELF CARE_CATEGORY_NOT_ALLOWED</code>	Self-care attempted blocked category.
<code>ASR_INFO_TEMPLATE_NOT_FOUND</code>	Answer template missing.
<code>ASR_INFO_TEMPLATE_VARIABLE_MISSING</code>	Required template variable not available.
<code>ASR_INFO_ANSWER_REQUIRED</code>	Cannot resolve without customer-visible answer.
<code>ASR_INFO_RESEARCH_TASK_OPEN</code>	Cannot resolve while required research is open.
<code>ASR_INFO_CONVERSION_TARGET_INVALID</code>	Target flow/category not allowed.

14. Minimum Implementation Order

Build in this order:

1. Seed TCK information request categories.
2. Add ASR info config migrations.
3. Seed answer templates.
4. Implement guidance API.
5. Implement self-care category validation.
6. Implement research task create/complete APIs.
7. Implement answer-and-resolve API.
8. Implement conversion/link API.
9. Emit ASR-02 business events.
10. Project to CUST-INT-01.

11. Patch Backoffice and self-care screens.

12. Add E2E tests.

15. Test Scenarios

15.1 Package Question Answered Immediately

Input:

- Self-care creates PACKAGE_EXPLANATION.

Expected:

- TCK ticket created.
- Guidance returns SIP source and template.
- CSR answers and resolves.
- Customer-visible answer appears in ticket/timeline.

15.2 Invoice Explanation Needs Research

Input:

- Customer asks about invoice line.

Expected:

- Research task created for billing-care.
- Ticket moves WAITING_INTERNAL.
- Completed research allows answer.

15.3 Missing Answer Blocks Closure

Input:

- User tries to resolve without answer body.

Expected:

- API returns ASR_INFO_ANSWER_REQUIRED.

15.4 Convert To Complaint

Input:

- Customer rejects explanation and requests correction.

Expected:

- Original info ticket is linked to ASR-03 complaint ticket or converted by policy.
- History preserved.

16. Developer Checklist

Before marking ASR-02 complete:

- Information request uses TCK ticket lifecycle.
- ASR-02 does not mutate business records.
- Guidance is config-driven.
- Self-care categories are explicitly allowed.
- Research tasks preserve source module and result.
- Final answer is customer-visible and stored.

- Conversion preserves original history.
- Customer timeline receives events.